

PART 2 :

I. POLYMERIZATION TECHNIQUES

II. CHEMICAL REACTION OF POLYMERS

III. BASIC PROPERTIES OF POLYMERS

- ▶ SOLUBILITY OF POLYMERS
 - ▶ SOLID STATE
 - ▶ THERMAL TRANSITION OF POLYMER
-

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I. Polymerization Techniques

I.I POLYMERIZATION CONDITIONS

□ Homogeneous system

All composition in the system (monomer, comonomer, solvent initiator or catalyst and so on) including polymer occurred are each other soluble through the synthesis performed. There are no phase separation during process. The system appear only one phase.

↳ bulk polymerization, solution polymerization

I.I POLYMERIZATION CONDITIONS

□ Heterogeneous system

The system consist of 2 phases. The reactant (monomer comonomer, initiator or catalyst, solvent) and product are insoluble in each other. It appear phase separation which it could be in some step or all step during synthesis.

↳ suspension polymerization, emulsion polymerization
: completely heterogeneous

↳ precipitation polymerization (partial heterogeneous), interfacial polymerization

I.2 FREE RADICAL POLYMERIZATION SYSTEM

I.2.1 Bulk or Mass polymerization

I.2.2 Solution polymerization

I.2.3 Suspension polymerization

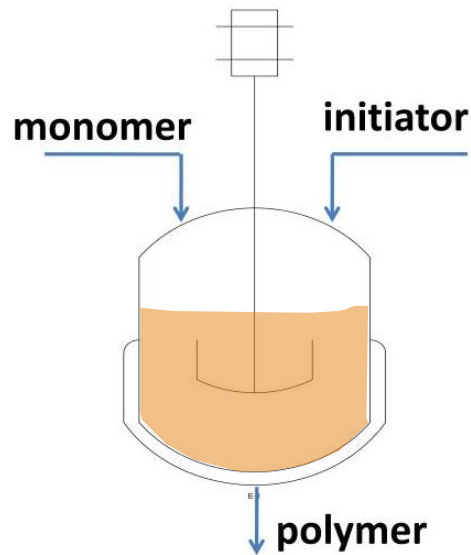
I.2.4 Emulsion polymerization

CONDENSATION POLYMERIZATION SYSTEM

I.2.5 Interfacial polymerization

1.2.1 Bulk or Mass polymerization

Only monomer (or comonomer) and a monomer-soluble initiator (catalyst) are used, no solvent or other chemicals.



Monomer (*liquid*)

condition:

initiator or catalyst (soluble in monomer!!!)

T, P

Polymer (*solid*)

- monomer-soluble polymer

“homogeneous sysm”

- monomer-insoluble polymer

“heterogeneous sysm”

Cite: <https://www.slideserve.com/rachel/polymerization-process>

Advantages of bulk polymerization

- ✓ simplest technique
- ✓ Give the high-purity polymer
- ✓ High yield per reactor volume
- ✓ Easy polymer recovery
- ✓ An option for casting the polymerization mixture into final product form

Problem in bulk technique

- ➔ Difficult to remove monomer residue because final product is solid form
- ➔ Problem of dissipating heat produced during polymerization, extremely high viscosity
- ➔ Low overall heat transfer coefficient

Free radical polymerization are highly exothermic reaction.
(~15-20kcal/kg)

↓
Reaction mixture has high temperature
and heat transfer is very difficult.

↓
Temperature in system could be more than 100°C

↳ Initiators and active species are decompose.

↳ Polymer degradation

↳ broad molecular weight distribution

Why it is difficult to transfer the heat?

$$R_p = k[M][I]^{1/2}$$

System contains only monomer and initiator

high [M] → high polymn rate (R_p)

heat from exothermic reaction
cause “Autoacceleration” or
“gel effect”

An increase in
temperature will
increase more the
polymzn rate.

Mol. Wt of polymer increase rapidly

High viscosity limit the heat diffusion ⇒ Inefficient stirring

kinetic of reaction change
from chemical controlled to a
diffusion controlled one

Heat removal is become difficult, : Explosion!!!!



broad molecular weight distribution

Monomer	Heat of Polymerization, Btu/lb ^a
Ethylene	1530–1660 (gas to solid)
Propylene	860
Isobutylene	370
Butadiene	620 (1,4 addition)
Vinyl chloride	650
Vinyl acetate	445 (at 170°F)
Acrylic acid	400 (at 166°F)
Ethyl acrylate	340
Methyl methacrylate	245
Styrene	290
Isoprene	470
Vinylidene chloride	330
Acrylonitrile	620

Cite: <https://www.slideserve.com/rachel/polymerization-process>

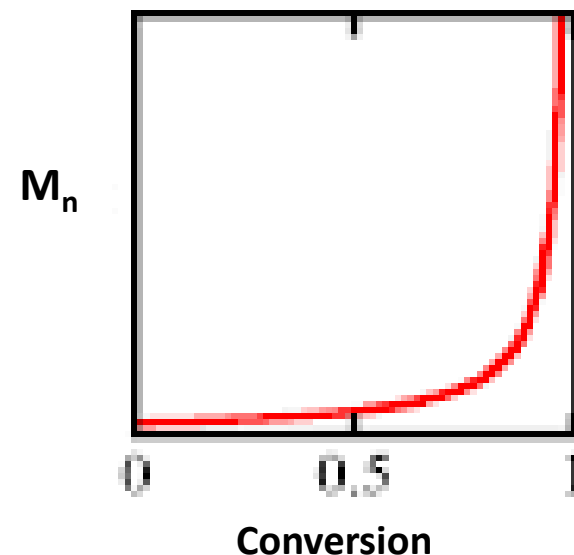
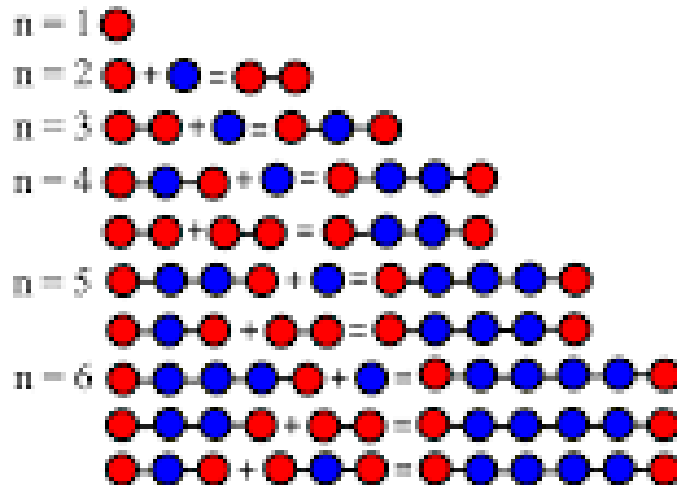
1 BTU (British Thermal Unit) คือ ปริมาณความร้อนที่ทำให้ น้ำ 1 ปอนด์ (0.45 กิโลกรัม) มีอุณหภูมิเพิ่มขึ้นหรือลดลง 1 องศาฟาเรนไฮต์ (0.56 องศาเซลเซียส)

NOTE

Bulk polymerization is generally used for step-growth polymerization (condensation) because :

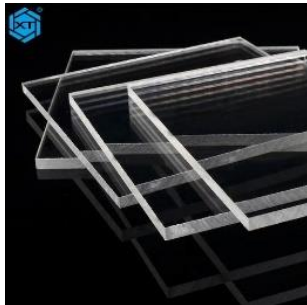
- condensation reaction is low exothermic reaction (2-6 Kcal/kg) and normally performed at high temperature ($> 100^{\circ}\text{C}$ - 200°C).
- There are no problem of dissipating heat because this reaction get the high molecular weight at nearly the end of reaction.

Step-growth



► However, bulk polymerization of some vinyl monomer are used, for example;

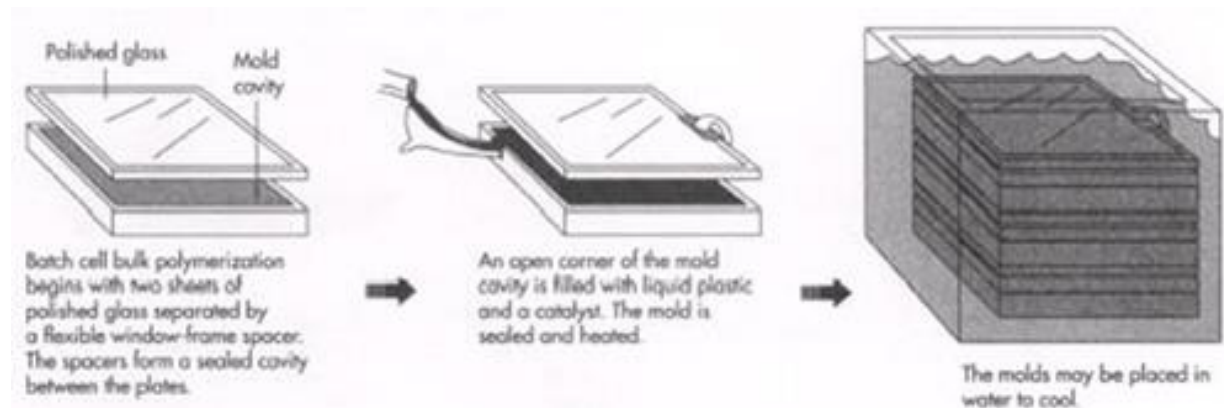
- preparation casting of PMMA, and PS (in sheet, rod, slabs);
- low MW polymer for use as adhesives, plasticizers, tackifiers and lubricant.
- preparation in small quantity and low mol.wt polymer.



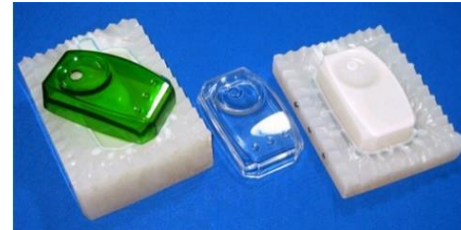
► To overcome the heat transfer problem, process is carried out in 2 stages or more :

Stage I: (Pre)polymerization in short period ➡ low MW polymer
(low viscosity)

Stage 2: Completely polymerization, often carried out in a mold resulting shaped products

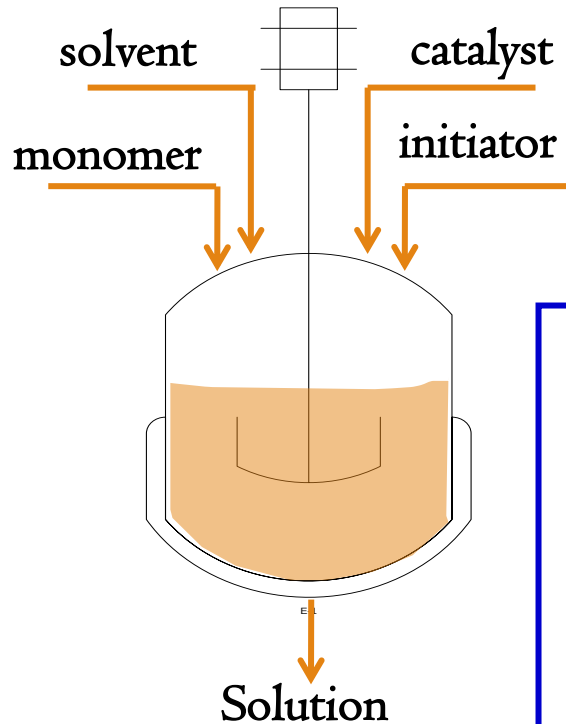


<http://atozplastics.com/upload/Literature/Batch-cell-and-continuous-bulk-polymerization-for-making-PMMA%20sheets.asp>



1.2.2 Solution polymerization

The system consists of monomers (comonomer), initiator (catalyst) are dissolved in solvent. (*normally is inert solvent e.g. toluene, chloroform, benzene, hydrocarbon*)



Solvent selection criteria :

- can dissolve monomers and initiator or other components.
- Low toxicity
- Suitable boiling point, easily removed.
- Inert to chain transfer reaction

Monomer/solvent

condition:

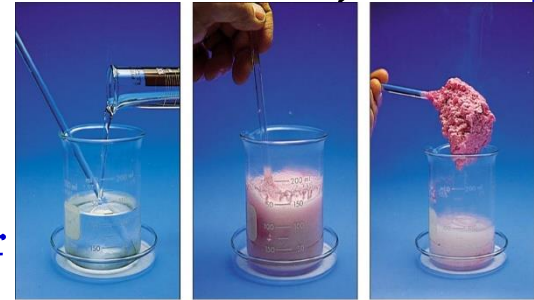
initiator or catalyst (dissolve in solvent!!!)

T, P

polymer

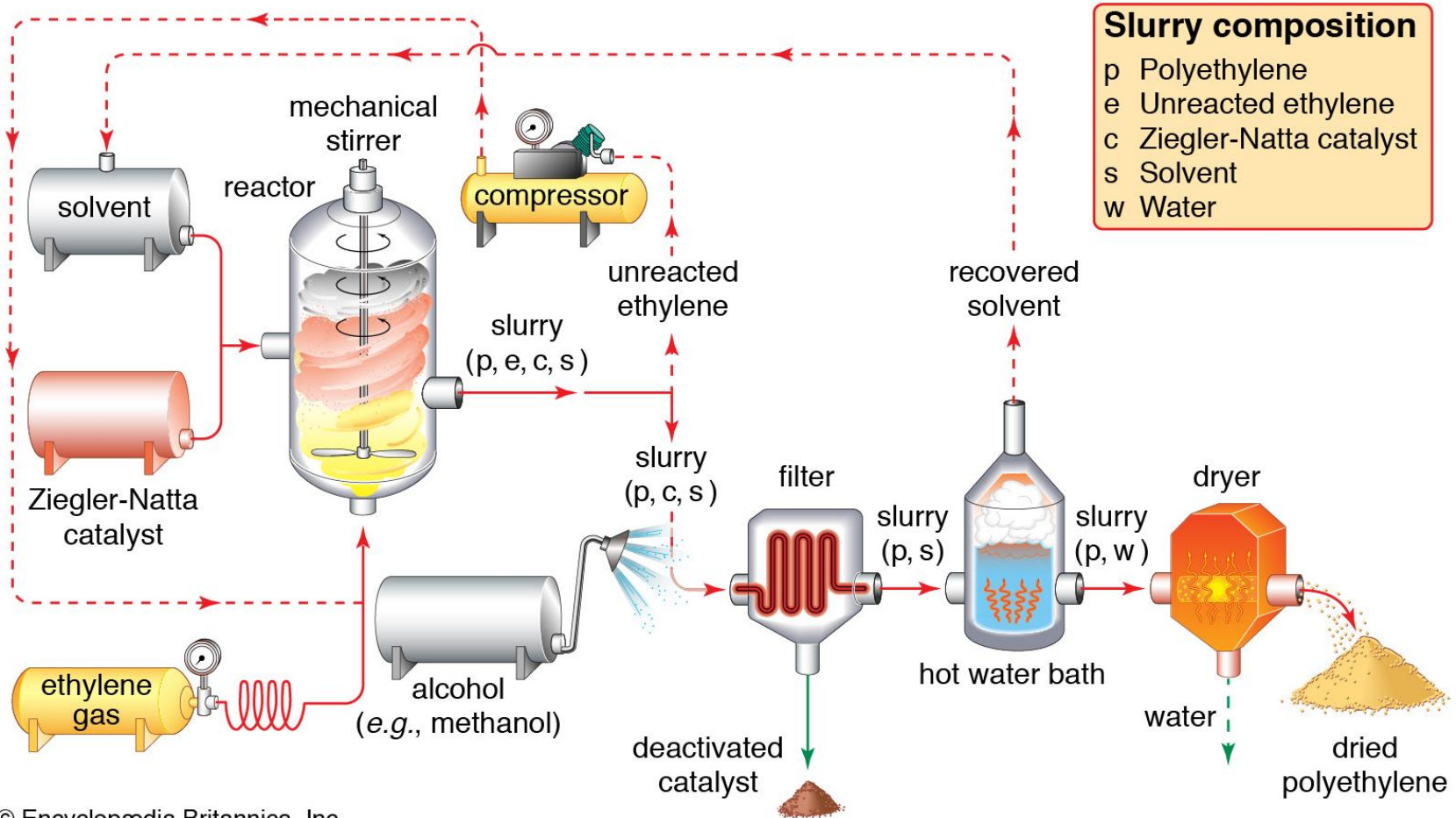
- solvent-soluble polymer
“homogeneous sysm”
- solvent-insoluble polymer
“heterogeneous sysm”

(precipitation polymerization)



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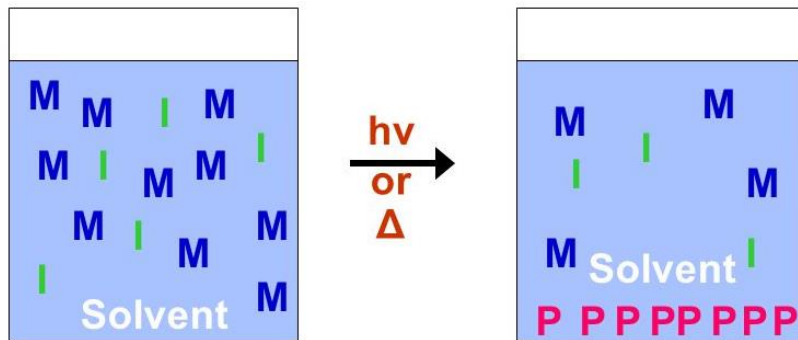
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Cite: <https://www.britannica.com/science/solution-polymerization>

Polymerization is performed in batchwise or continuously solution polymerization

Precipitation polymerization

a heterogeneous polymerization process that begins initially as a homogeneous system where the monomer and initiator are completely soluble, but upon initiation the formed polymer is insoluble and thus



- Polymer precipitation depends on MW , crystallinity, rate of polymerization
- Polymerization is continues
- Used for :
 - Polyacrylonitrile in water
 - Fluoroolefins in CO₂
 - Poly(acrylic acid) in benzene
 - Poly(acrylic acid) in CO₂

Consideration:

- Ease of separation
- *Traditionally, not applicable..*
(polymer must be insoluble in its own monomers)



Advantages of solution polymerization

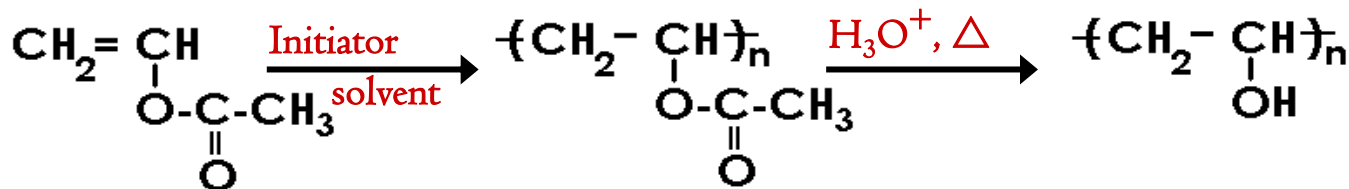
▷ Monomer + initiator (or catalyst) + solvent

Low viscosity
allowing efficient heat transfer

▷ The catalyst can be removed from polymer

▷ It can use for synthesis the polymers that can not be prepared directly from their monomer such as poly(vinyl alcohol) which synthesized from hydrolysis of poly(vinyl acetate) (PVAc) after synthesize PVAc from vinyl acetate.

Or in case there are chemical modification after polymerization.





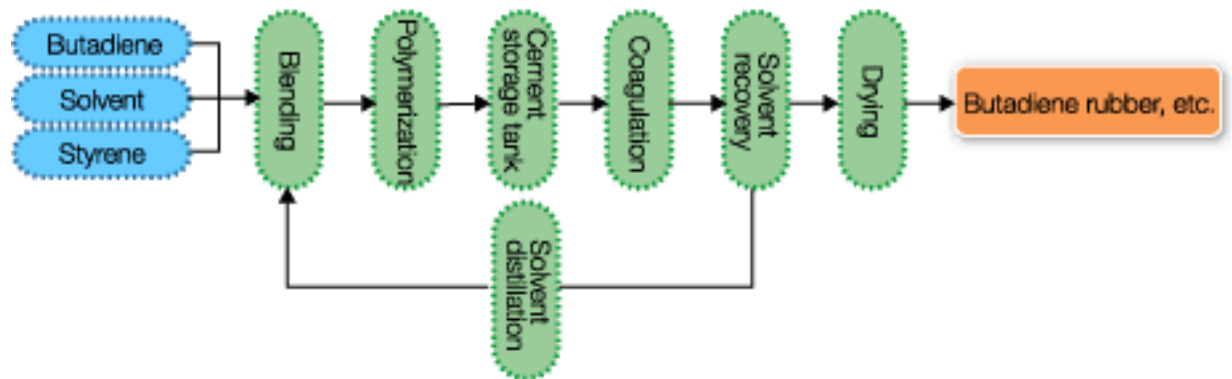
Drawback of solution polymerization

- ▷ The solubility of polymer is limited at high MW.
- ▷ Low reaction rate and degree of polymerization
- ▷ Chain transfer to solvent could occur $\Rightarrow$ result in limit MW of polymer
- ▷ Necessity of selecting an inert solvent
- ▷ Need solvent removal process and it maybe difficult to completely remove.
- ▷ Costly and toxicity of solvent.



Commercial use

- ▷ Adhesives and coating processes (poly(vinyl acetate) $\rightarrow$ poly(vinyl alcohol), and some acrylic ester
- ▷ Use widely in ionic and coordination polymerization
- ▷ High-density polyethylene, polybutadiene, and butyl rubber production



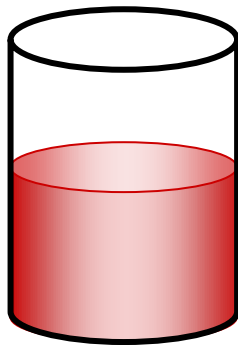
Pure polymer and
no heat dissipating problems



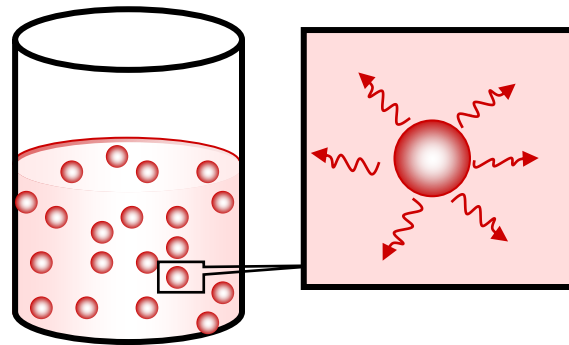
Suspension poly^m
Emulsion poly^m

Heterogeneous system

► Heterogeneous system (high surface area) is used extensively
to control the thermal and viscosity problem.



Homogeneous system



Heterogeneous system

1.2.3 Suspension polymerization

The system consist of monomer, comonomer and initiator dispersed in medium (i.e. H_2O). Monomers and initiator is dissolve each other.

They are disperse in droplet form in medium.

Batch reactor fitted with mechanical agitator is charged with a water-insoluble monomer and initiator.

Monomer/solvent

condition:

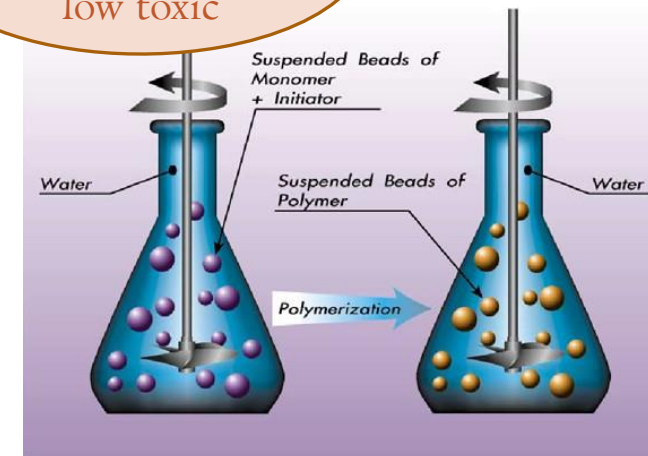
initiator or catalyst (dissolve in monomers!!!)

T, P

suspension agent + medium (H_2O)

polymer bead (or pearl) in medium
“heterogeneous sysm”

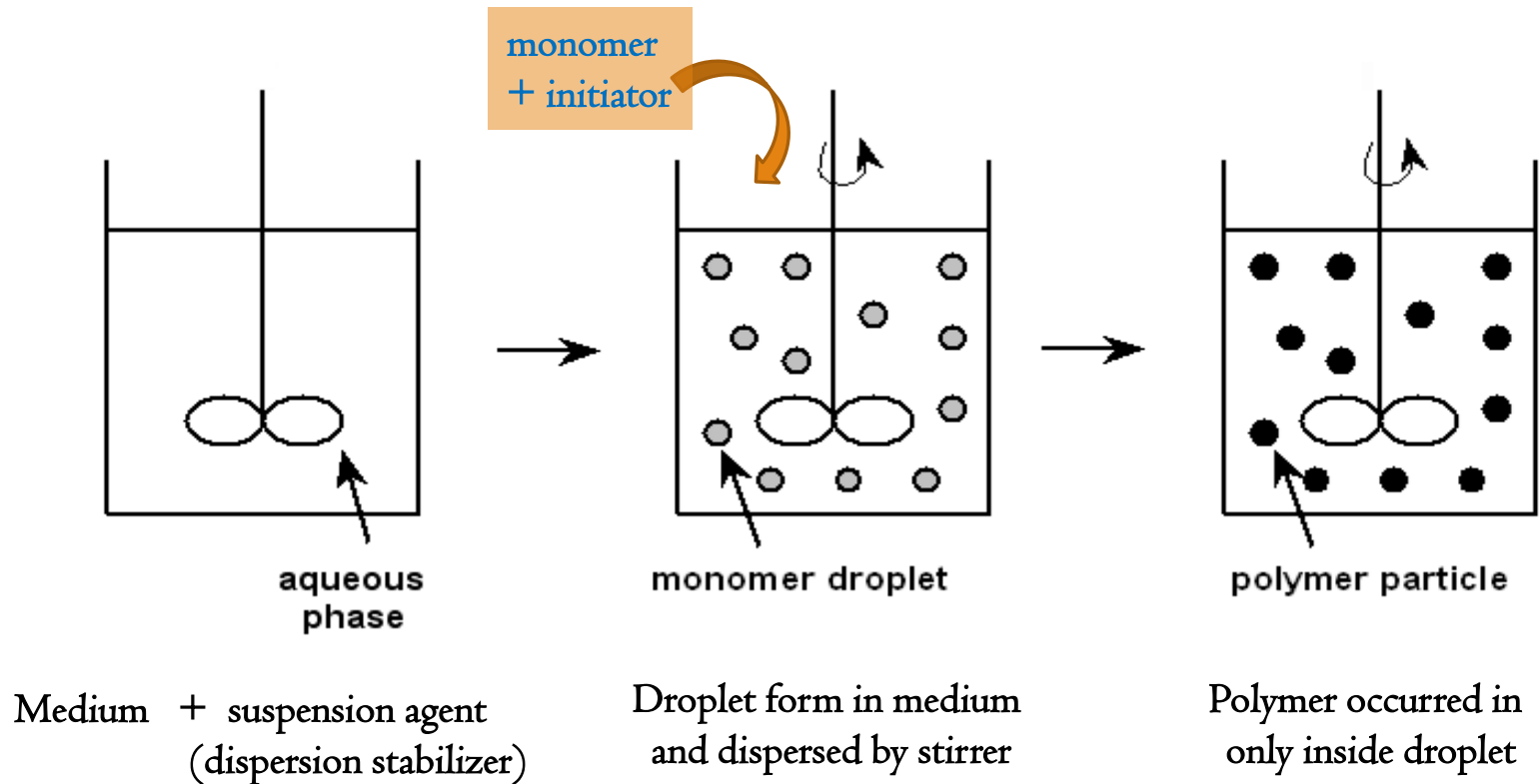
High thermal
conductivity,
low toxic



Dispersion (suspension) stabilizer = dispersant = suspension agent

1.2.3 Suspension polymerization

Polymerization procedure:

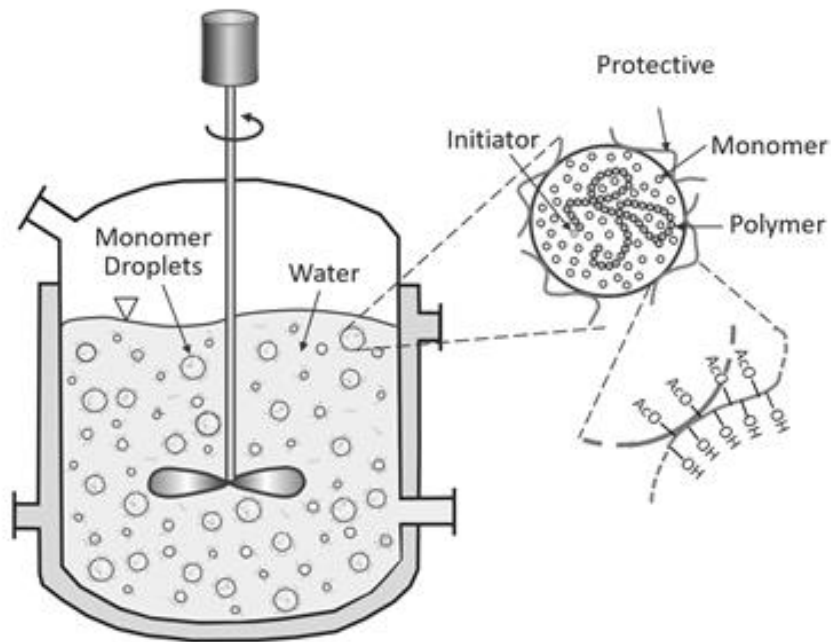


Stabilizer : water soluble polymers eg. Poly(vinyl alcohol), Methyl cellulose, sodium polyacrylate, gelatin

Cite: <http://chemicalizereng.blogspot.com/2011/06/suspension-polymerization.html>

I.2.3 Suspension polymerization

❖ Droplet of monomer containing the initiator are formed, typically 50-200 μm in diameter and serve as *mini-reactor of polymerization*.



1 droplet = 1 bulk reactor



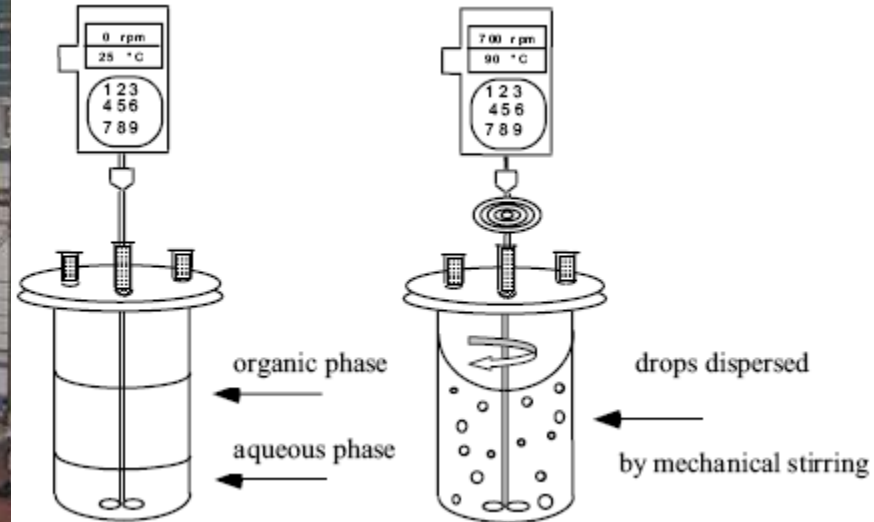
mini-bulk polymerization

❖ Droplet composition and reaction are the same as bulk polymerization; it is only small and dispersed separately in medium.

❖ Stabilizer is a **protective suspension** that prevents coalescence of monomer droplets and sticky polymer particles.

❖ Medium can transfer the heat; surface area of droplet is high, so there is no problem about heat dissipation.

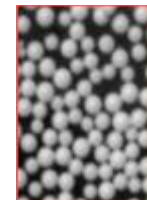
Also called « Pearl or bead polymerization »



I.2.3 Suspension polymerization

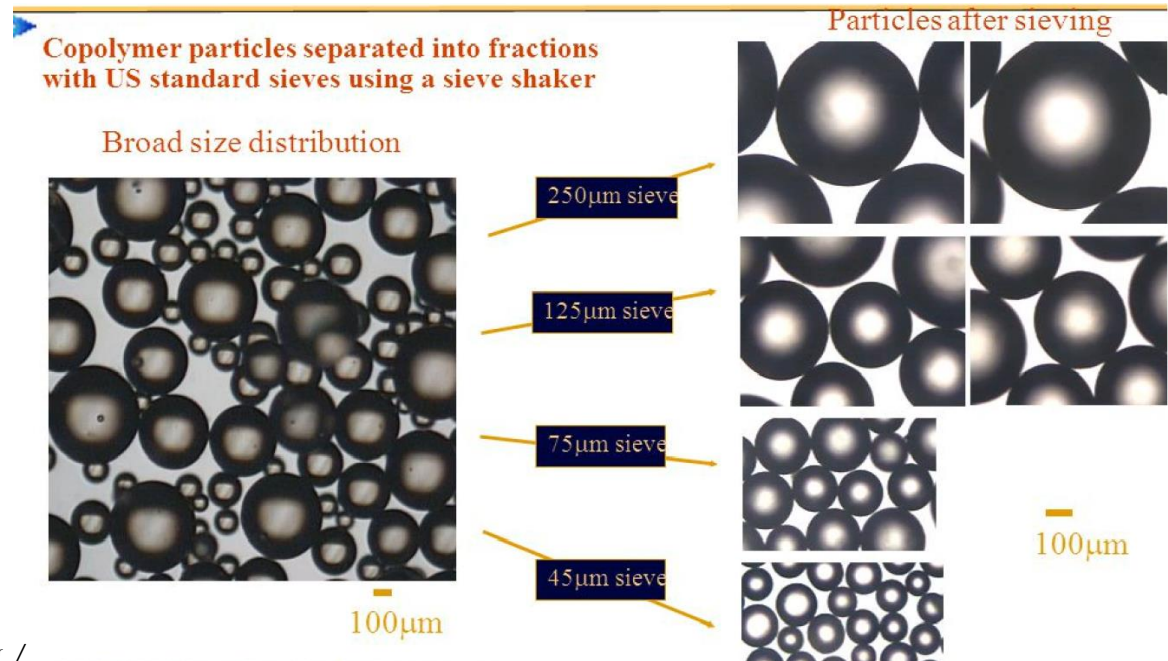
Characteristic of suspension polymerization

- Initiator & monomer: insoluble in medium
- Monomer (contained initiator) *suspended* in liquid phase such as H₂O in *droplet form*. (*mini-bulk polymerization*)
- Monomer droplet formed by stirring and stirring speed affects polymer particle size.
- Polymerization occurs in monomer droplets.
- Monomer droplet is stabilized by “suspension or dispersion stabilizer,” and it protects the polymer particles from sticking together.
- Various particle size product (50-200 μm)



Advantages of suspension polymerization

- Cheap, as it involves only water instead of solvents
- Viscosity increases is negligible
- Agitation and temperature is easily controlled.
- Particle size control by speed of stirrer.
- Products is easily recovered by filtration and directly used in bead form



Disadvantages of suspension polymerization

- It can be adopted only for water-insoluble monomers.

Inversion-suspension must be applied for soluble monomers

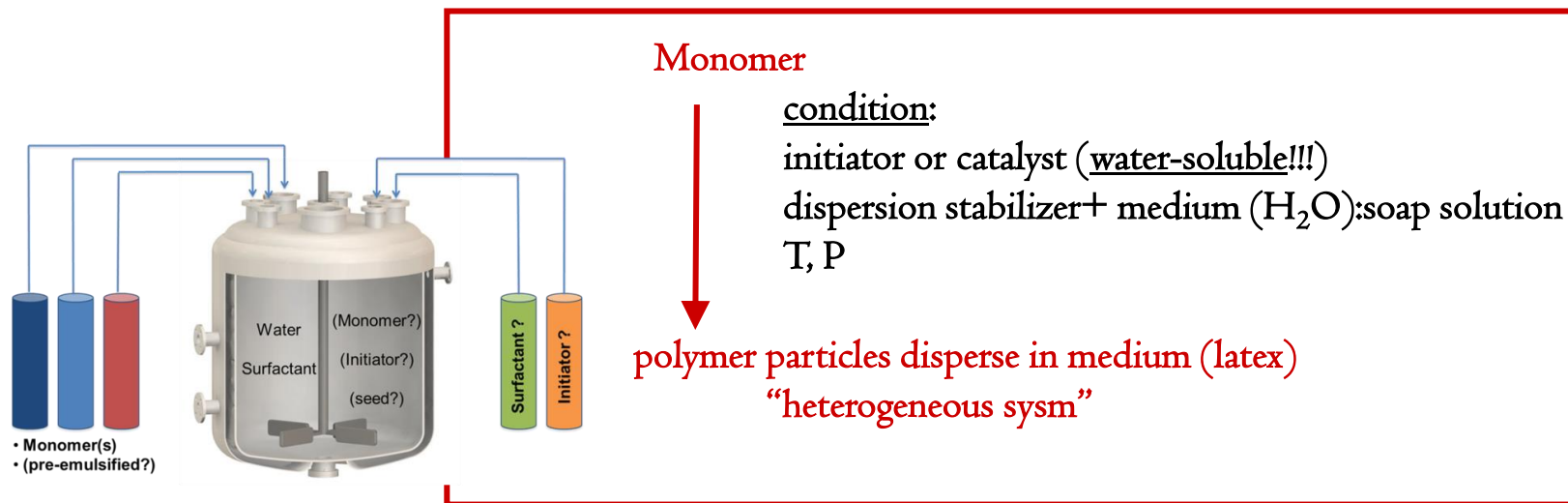
- Low purity, difficult to remove all stabilizers and other additives
- Large volume of reactor is taken up by water
- Can't be used for tacky polymers such as elastomer (tendency for agglomeration)

1.2.4 Emulsion polymerization

Utilize **water** as **heat transfer agent**.

Difference from suspension polymerization :

- use **water-soluble initiator** (insoluble in monomer).
- Much **smaller particle size** than suspension plmzn., in **colloidal form** called latex



dispersion stabilizer = emulsifying agent = emulsifier

I.2.4 Emulsion polymerization

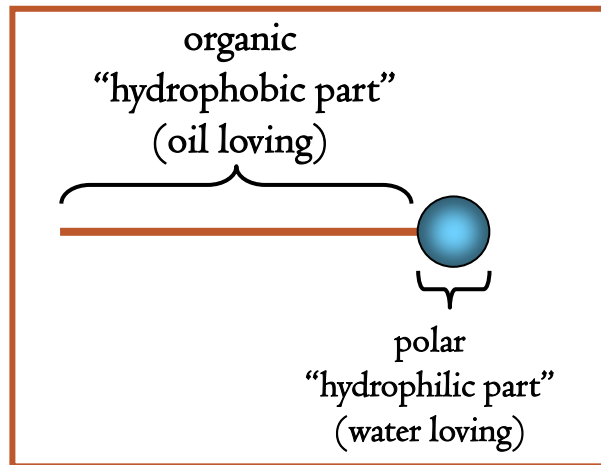
Typical Emulsion polymerization Recipes of styrene-butadien rubber

Ingredients	%wt
Butadiene	75
Styrene	25
Water	180
Soap	5.0
Lauryl mercaptan	0.50
Potassium persulfate	0.30

► dispersion stabilizer used: anionic surfactant

ex. sodium lauryl sulphate: $\text{H}_3\text{C}(\text{CH}_2)_{11}\text{SO}_4^- \text{Na}^+$
sodium stearate : $\text{H}_3\text{C}(\text{CH}_2)_{16}\text{COO}^- \text{Na}^+$

► The surfactant molecules, composed of a hydrophilic (water-attracting) and hydrophobic (water-repelling) end.

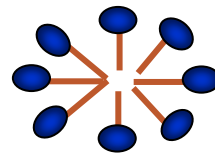


There are micelle formation.
Monomer droplets are stabilize by micelle.

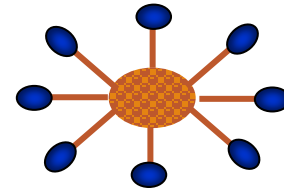
1.2.4 Emulsion polymerization

► Micelle formation

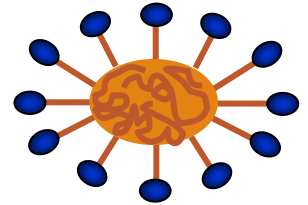
Micelle: an aggregate of surfactant molecules dispersed in liquid colloid, e.g. in aq. solution, hydrophilic part (head) contacts with water, and hydrophobic part is in the center.



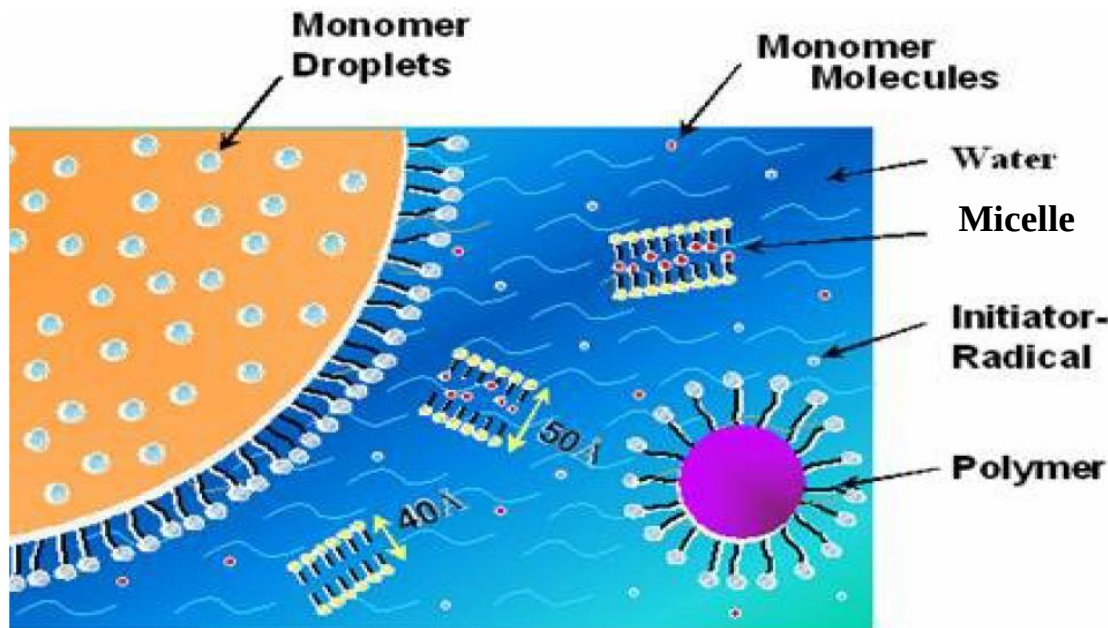
micelle
(disc shaped)
40-50 Å



monomer-swollen micelle
(5- 6.5 nm)

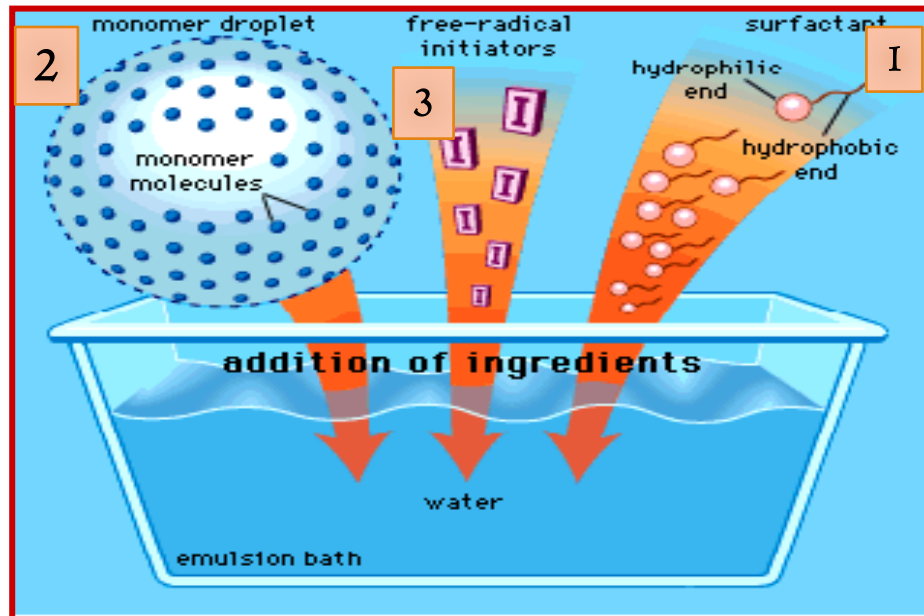


polymer particle
(100 nm)



I.2.4 Emulsion polymerization

- I. Monomer molecules and FR initiators are added to a water-based emulsion bath along with surfactants, or soap.

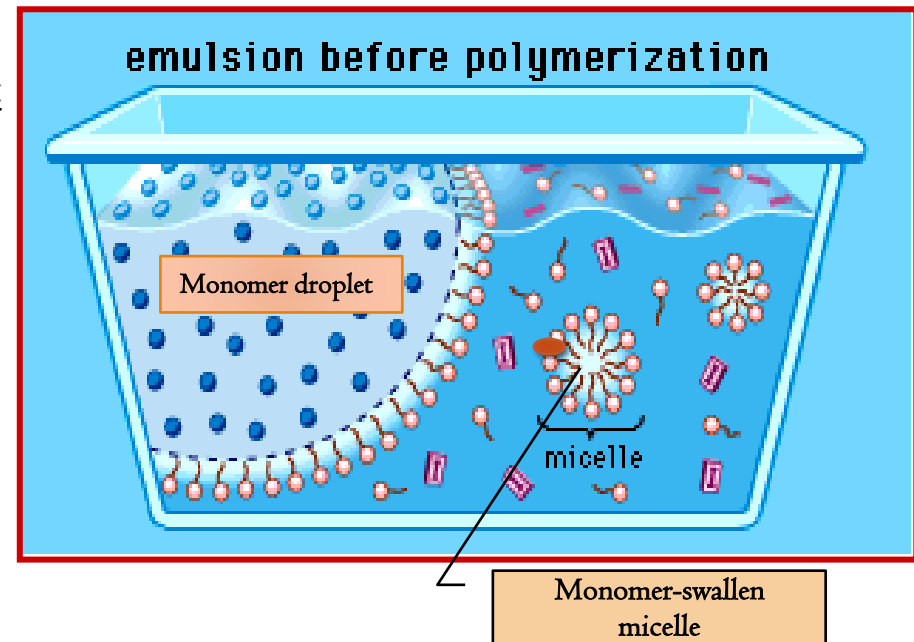


1.2.4 Emulsion polymerization

2. Before polymerization, the surfactant molecules form a **stabilizing emulsion** by coating monomer droplets. Other surfactant molecules clump together into smaller aggregates called **micelles**, which also absorb monomer molecules.

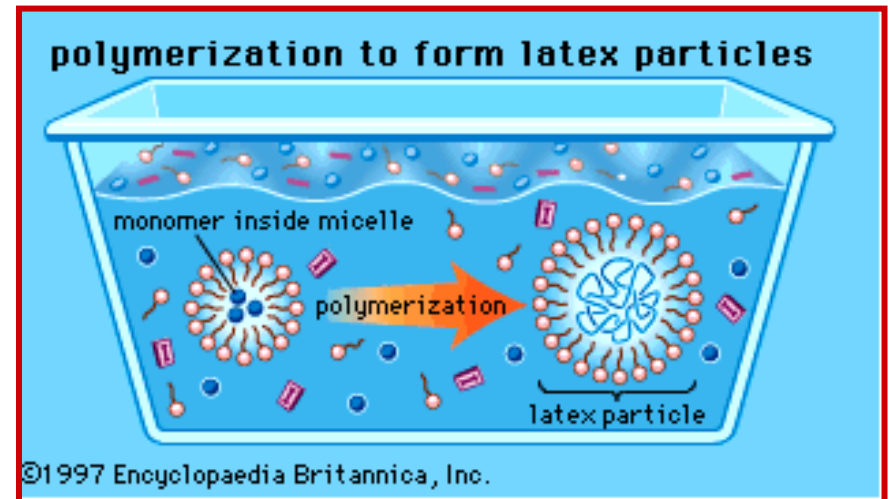
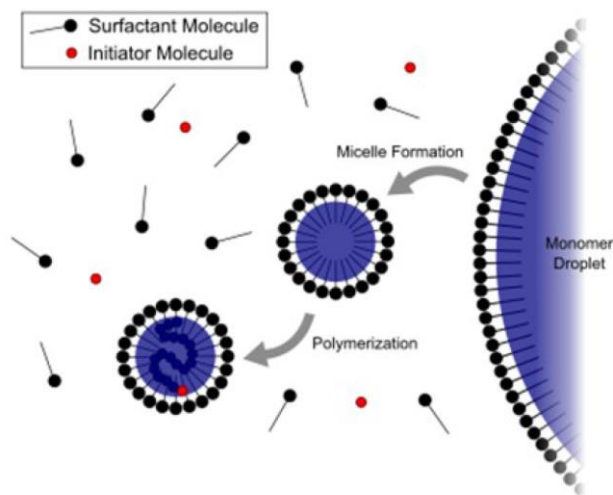
Before Polymerization, it consisted of 3 compositions:

1. Medium (H_2O) + initiator + some soluble monomers
2. Monomer droplets are surrounded by surfactant
3. Monomer-swollen micelles (much smaller than monomer droplet)



1.2.4 Emulsion polymerization

3. Polymerization occurs when **initiators migrated into the micelles**, induces monomer molecules to form polymer that makes up the latex particles.



Polymerization occur in monomer-swollen micelles. There are a lot of micelles, and they are small → polymer is small particles.

I.2.4 Emulsion polymerization

Characteristic of emulsion polymzn

- initiator **soluble in medium, insoluble in monomer**
- monomer droplet is stabilized by **surfactant** in micelle form and surfactant prevent an agglomerate of polymer particle.
- **Polymerization occur in micelles**
- **Termination is low**, chain transfer agent may be needed for control MW
- Polymer particle is very small. **Product is in latex form**
- Latex **can be directly used in some application**, e.g. paint, adhesive, synthetic rubber latex (styrene-butadiene rubber (SBR), Acrylonitrile-butadiene rubber (NBR))
- **low purity**, difficult to remove all stabilizer and other additives

I.2.4 Emulsion polymerization

Advantages

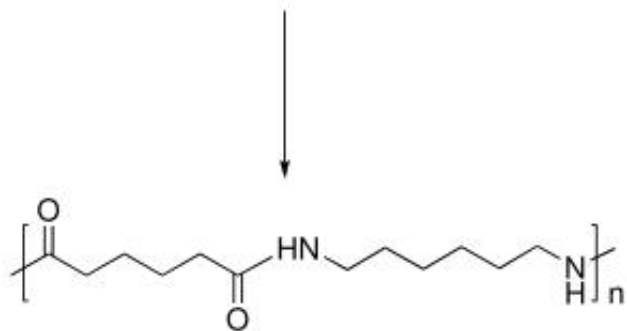
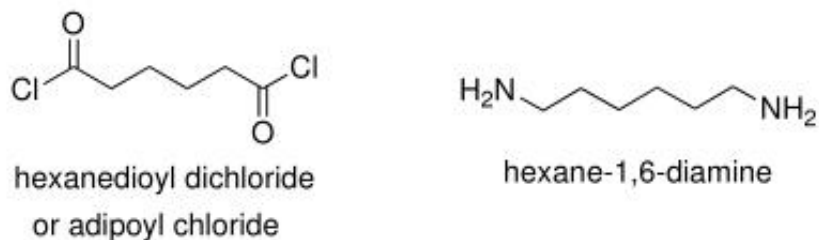
- High molecular weight polymers
- fast polymerization rates.
- allows removal of heat from the system.
- viscosity remains close to that of water and is not dependent on molecular weight.
- The final product can be used as such ,does not need to be altered or processed

Disadvantages

- Surfactants and polymerization adjuvants -difficult to remove
- For dry (isolated) polymers, water removal is an energy-intensive process
- Designed to operate at high conversion of monomer to polymer. This can result in significant chain transfer to polymer.
- Can not be used for condensation, ionic or Ziegler-Natta polymerization.

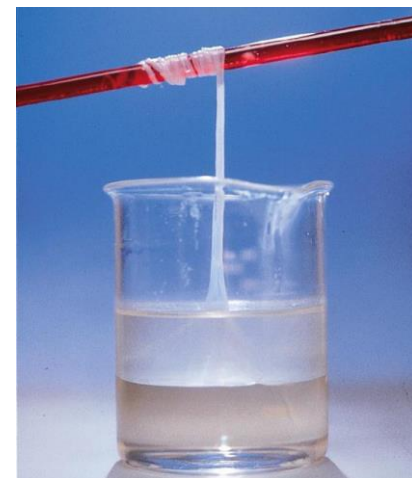
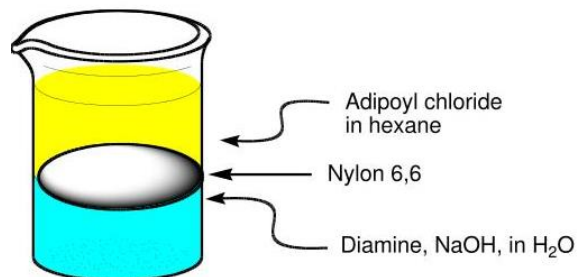
3.2.5 Interfacial polymerization

Polymerization of two reactants performed at the interface between two phases, each containing one of the reactant, normally at **room temp.**



Nylon 6,6

Use in polyamide (or nylon synthesis)
using diamine ad



The reactant diffuse to interphase and **undergo polymerization** at the interface.

The polymer product precipitates and is **continously withdraw** in form of continuous film or filament.

References

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